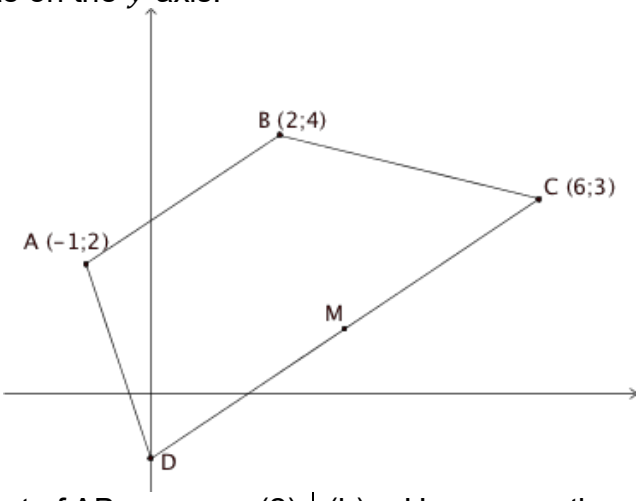
Page a day worksheet **PAD nr: 15**
Grade 10 – Analytical Geometry

Name:	

In the following diagram we have A (-1;2), B(2;4), C(6;3) and D $(x; y)$ forming a trapezium with AB∥CD and D a coordinate on the y-axis.



(a) Determine the gradient of AB. (2)

(b) Hence, or otherwise, determine the coordinates of D. (3)

(c) If M is the midpoint of CD and D's y-coordinate -1, determine M's coordinates. (2)

(d) Is ABCD an isosceles trapezium? Show all calculations to justify your statement. (4)

(e) Determine the equation of AB. (2)

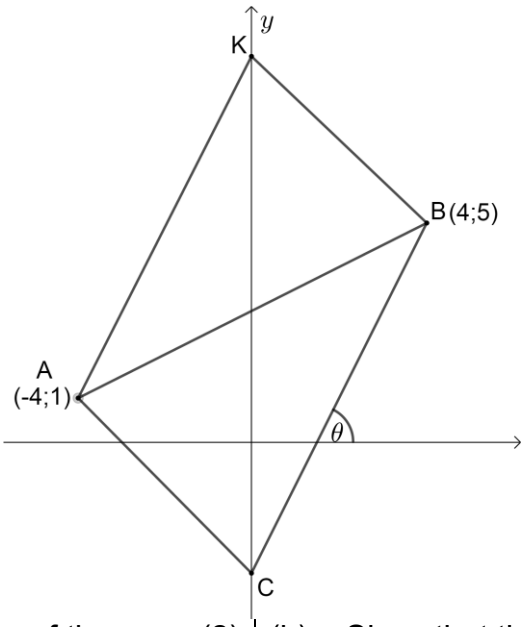
(f) Is $AB \perp CD$? Show all calculations. (3)

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Page a day worksheet **PAD nr: 16**
Grade 10 – Analytical Geometry

Name:	

The diagram below shows quadrilateral AKBC, with A (-4;1) and B (4;5). C is on the y-axis and K is the fourth vertex of the quadrilateral. The gradient of the line segment BC is 2.



(a) Determine the coordinates of the midpoint of AB. (2)

(b) Show that the coordinates of C are (0;-3). (3)

(c) Determine K's coordinates such that AKBC is a parallelogram. (2)

(d) Determine the equation of AB. (4)

(e) Will A, B and M (8;9) be collinear? Show all calculations to justify your answer. (4)

(f) Is AC⊥BC? Show all calculations. (3)

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